ECEN 5613 FINAL PROJECT · BUILD PLAN · REV 7 · 2026-10-18

MIDI Foot Controller for Reaper

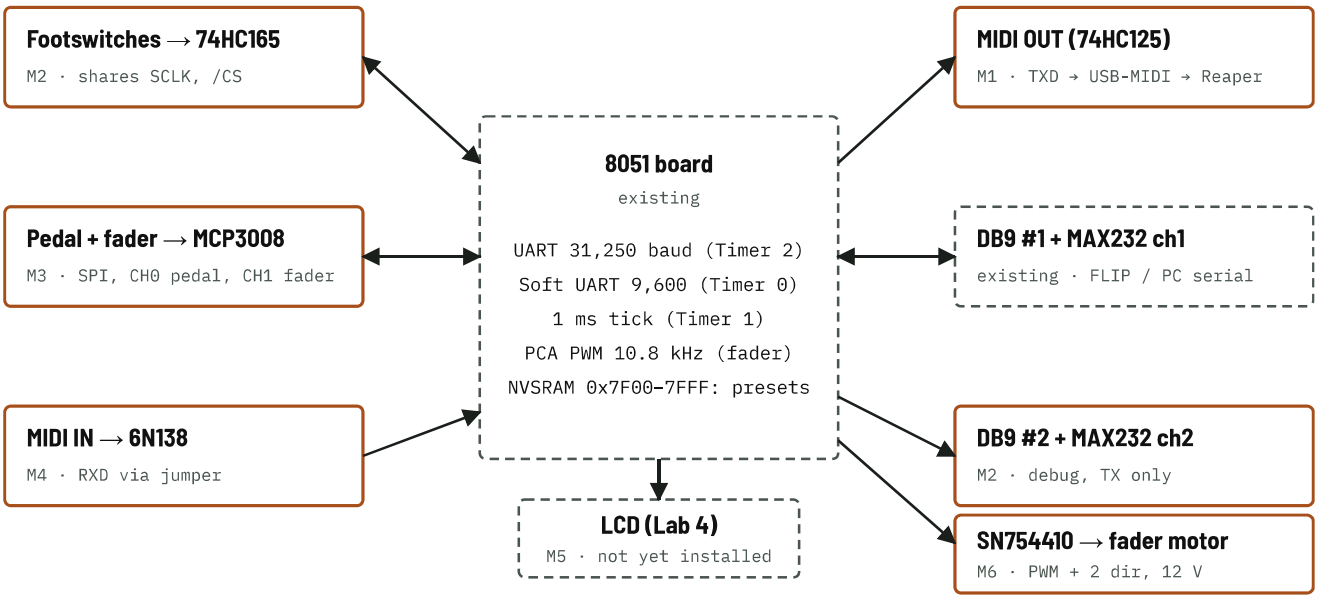
8051-based MIDI controller. Inputs: 4 footswitches, expression pedal, motorized fader, MIDI IN. Outputs: MIDI OUT to Reaper (via USB-MIDI), fader motor, LCD status, debug serial. Seven milestones, each independently demonstrable.

> **Scope of this document:** team build plan. Not the PDR submission. PDR deliverable: 3–6 slides, file `F26_FirstLast_FirstLast.pptx` / `.pdf`, footer on every slide (project name, both names, slide no., date). Due on Canvas Sun 10/18.

System overview

MIDI 1.0: asynchronous serial, 31,250 baud, 8-N-1, 5 mA current loop. New hardware is interface circuitry between the 8051 port pins and each connector or actuator.

New hardware Existing / on hand



Constraint: one hardware UART, already connected to MAX232 ch1 for FLIP and PC serial. Resolution: MIDI OUT shares TXD (fan-out, no conflict). MIDI IN shares RXD through a 3-pin jumper (RUN / PC). Debug output moves to a TX-only software UART on P3.5 → MAX232 ch2 → DB9 #2.

Board baseline

Source: board schematic rev 5 (2026-10-16), Labs 1–3 core elements. Supplemental elements not implemented.

RESOURCE	CURRENT STATE	IMPACT ON PLAN
MCU	AT89C51RC2, 11.0592 MHz, X1 mode.	Timer 2 and PCA available. All reload values assume X1.
P1.1 (pin 2)	**In use:** status LED via 2.2 kΩ → PN2907A.	Not available.
P1.0, P1.2–P1.7	Unconnected.	SPI, fader PWM/direction, switch data.
P3.0 RXD / P3.1 TXD	MAX232 R1OUT (pin 12) → RXD; TXD → T1IN (pin 11).	RXD rework in M4. TXD shared with MIDI OUT.
P3.2–P3.5	Unconnected.	/CS, soft UART TX, 2 reserved.
P3.6 /WR, P3.7 /RD, P0, P2	External bus: '373 latch, bq4011, SPLD. Port 0 pull-ups installed.	Unavailable.
MAX232 ch2 (pins 7–10)	Unconnected.	DB9 #2.
bq4011 NVSRAM	XRAM 0x0400–0x7FFF (31 KB). Internal XRAM 0x0000–0x03FF.	Presets at 0x7F00–0x7FFF. Exclude that region from the SDCC XRAM allocation.
0x8000–0xFFFF data space	Unmapped. Lab 3 virtual debug port at 0xFFFF.	Logic-analyzer tracing available.
/PSEN, /EA	SW3 + 1 kΩ to GND on /PSEN (bootloader entry). J9 selects /EA.	FLIP requires RXD jumper in PC position.
LCD, serial EEPROM	Not installed. Lab 4 hardware.	M5 depends on Lab 4 LCD. P3.2/P3.3 held for Lab 4.

Pin allocation

PIN	DIP-40	FUNCTION	MILESTONE
P1.0	1	SCLK → MCP3008 pin 13 and 74HC165 pin 2 (shared)	M2, M3
P1.1	2	Status LED (existing)	—
P1.2	3	MOSI → MCP3008 pin 11	M3
P1.3	4	MISO ← MCP3008 pin 12	M3
P1.4	5	CEX1 (PCA PWM) → SN754410 pin 1 (1,2EN)	M6
P1.5	6	DIR A → SN754410 pin 2 (1A)	M6
P1.6	7	DIR B → SN754410 pin 7 (2A)	M6
P1.7	8	Switch data ← 74HC165 pin 9 (QH)	M2
P3.0	10	RXD ← jumper (MIDI / PC)	M4
P3.1	11	TXD → MAX232 T1IN + 74HC125	M1
P3.2, P3.3	12, 13	Reserved (Lab 4)	—

PIN	DIP-40	FUNCTION	MILESTONE
P3.4	14	/CS → MCP3008 pin 10 and SH/LD → 74HC165 pin 1 (shared)	M2, M3
P3.5	15	Soft UART TX → MAX232 T2IN	M2

Pin budget: 11 free, 9 allocated, 2 reserved. Footswitches go through a 74HC165 shift register that shares SCLK and /CS with the ADC. This saves 2 pins versus direct wiring and frees P1.4–P1.6 for the fader. Fader PWM uses PCA output CEX1 (P1.4); P1.5/P1.6 are plain GPIO.

Bill of materials

Prices: single-quantity estimates (Digi-Key / Mouser). Order one spare of each IC. Socket all ICs.

PART	FUNCTION	QTY	≈ COST	USED IN
ICs				
74HC125 quad buffer, DIP-14 (SN74HC125N)	MIDI OUT line driver. Sinks 5 mA loop current; 8051 port V_{OL} is specified at only 1.6 mA I_{OL}.	2	$1	M1
74HC165 8-bit PISO shift register, DIP-16 (SN74HC165N)	Reads 4 footswitches (4 spare inputs) over 2 shared pins + 1 data pin.	2	$1.50	M2
MCP3008-I/P ADC, DIP-16	8-ch, 10-bit SPI ADC. CH0 pedal, CH1 fader position.	2	$7	M3
6N138 optocoupler, DIP-8	MIDI IN galvanic isolation (required by MIDI 1.0 spec).	2	$3	M4
SN754410NE quad half-H driver, DIP-16	Fader motor H-bridge. 1 A/ch continuous, 4.5–36 V motor supply, 5 V logic, internal clamp diodes.	2	$5	M6
IC sockets: DIP-8 ×1, DIP-14 ×1, DIP-16 ×3	Replaceable ICs.	—	$3	All
Connectors / switches				
5-pin DIN female, 180° (panel-mount solder-cup, or PCB e.g. CUI SDS-50J)	MIDI IN, MIDI OUT.	3	$5	M1, M4
DB9, gender matching DB9 #1	Debug port.	1	$2	M2
Footswitch, momentary SPST, soft-touch	User input.	4	$12	M2
1/4" TRS jack (any in stock; Switchcraft 112BX lead time 10 wk at Digi-Key)	Expression pedal input.	1	$4	M3
3-pin male header + shunt	RXD source select.	1	$1	M4
Passives				
Resistors ¼ W: 220 Ω ×5, 470 Ω ×2, 1 kΩ ×2, 4.7 kΩ ×2, 10 kΩ ×8	Current limit, pull-ups (74HC165 inputs have none), bias.	—	$1–8	All
1N4148	6N138 LED reverse protection.	2	$0.20	M4
0.1 µF ceramic	Decoupling, one per IC; ADC input filters.	10	$1.50	All
100 µF electrolytic, ≥ 25 V	Motor supply bulk capacitance at SN754410 pin 8.	2	$0.60	M6
10 kΩ linear pot	Bench substitute for pedal.	1	$2	M3

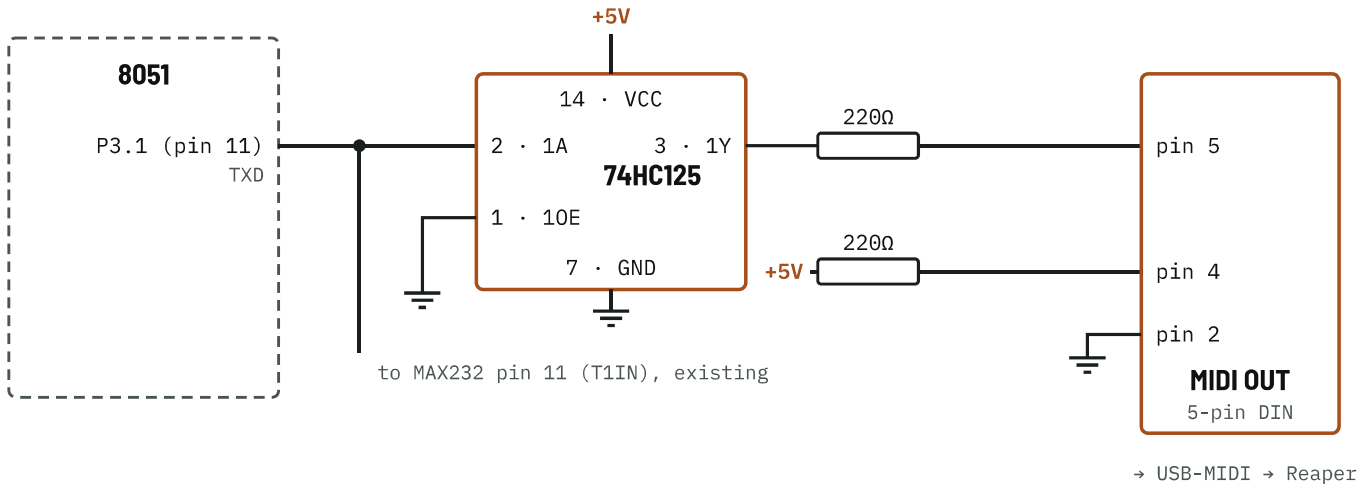
PART	FUNCTION	QTY	≈ COST	USED IN
Cables / test				
USB-MIDI interface, M-Audio Uno or Roland UM-ONE mk2	PC link. Integrated DIN plugs; no separate MIDI cables. Avoid unbranded cables (known data loss).	1	$35–45	M1
TRS expression pedal, M-Audio EX-P (optional)	Demo input. Pot is sufficient.	1	$30	M3
On hand				
✓ 8051 board (Labs 1–3), MAX232, NVSRAM, PSU, wire-wrap tools, Reaper. Course kits: 2× USB-RS232, USB-TTL, LogicPort analyzer. AE lead: SM100AN-B10K motorized faders, USB-PD trigger module (5/9/12/24 V), existing motor drivers. Software: loopMIDI, MIDI-OX, Hairless MIDI Serial Bridge (fallback).				

Total: $85–105 base. +$30 pedal → $115–135. Existing motor drivers may replace the SN754410 if rated ≥ 1 A continuous at 12 V with 5 V-compatible logic inputs.

Milestones

Sequential per owner. Next milestone starts after the current one passes its acceptance test. Pin numbers: 40-pin DIP.

M1 MIDI OUT `Owner: both`



HARDWARE

- 74HC125: pin 14 → +5V, pin 7 → GND, 0.1 µF pin 14–GND at the IC.
- Unused inputs (pins 4, 5, 9, 10, 12, 13) → GND.
- DIN pin order is non-sequential (1-4-2-5-3); rear view is mirrored. Verify against the connector drawing with a continuity check before soldering.

FIRMWARE

- UART initialization with Timer 2 as baud generator.
- Byte transmit; 3-byte Note On / Note Off messages. Example: `0x90 0x3C 0x64` = ch 1, C4, vel 100.
- Test loop: Note On, 500 ms, Note Off, repeat.

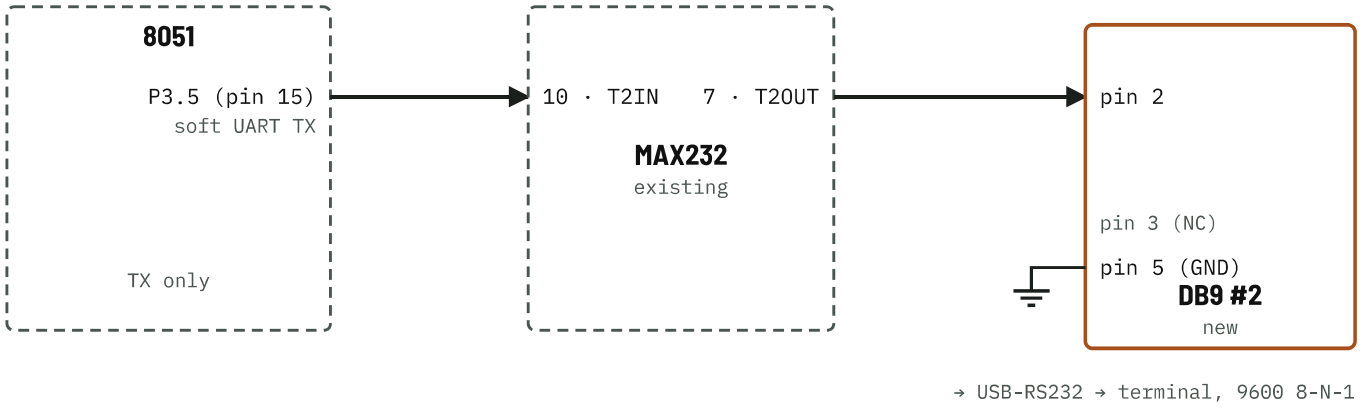
PARAMETER	VALUE
Baud formula (Timer 2, X1)	baud = F_{osc} / (32 × (65536 − RCAP2)), F_{osc} = 11.0592 MHz
Timer 2 reload, MIDI	0xFFF5 → 31,418 baud, +0.54% (MIDI tolerance ±1%)
UART mode	Mode 1 (8-N-1), receiver enabled, Timer 2 clocks both RX and TX
X2 mode reloads (if enabled)	T2 0xFFEA · T0 0x40 · T1 0xF8CD

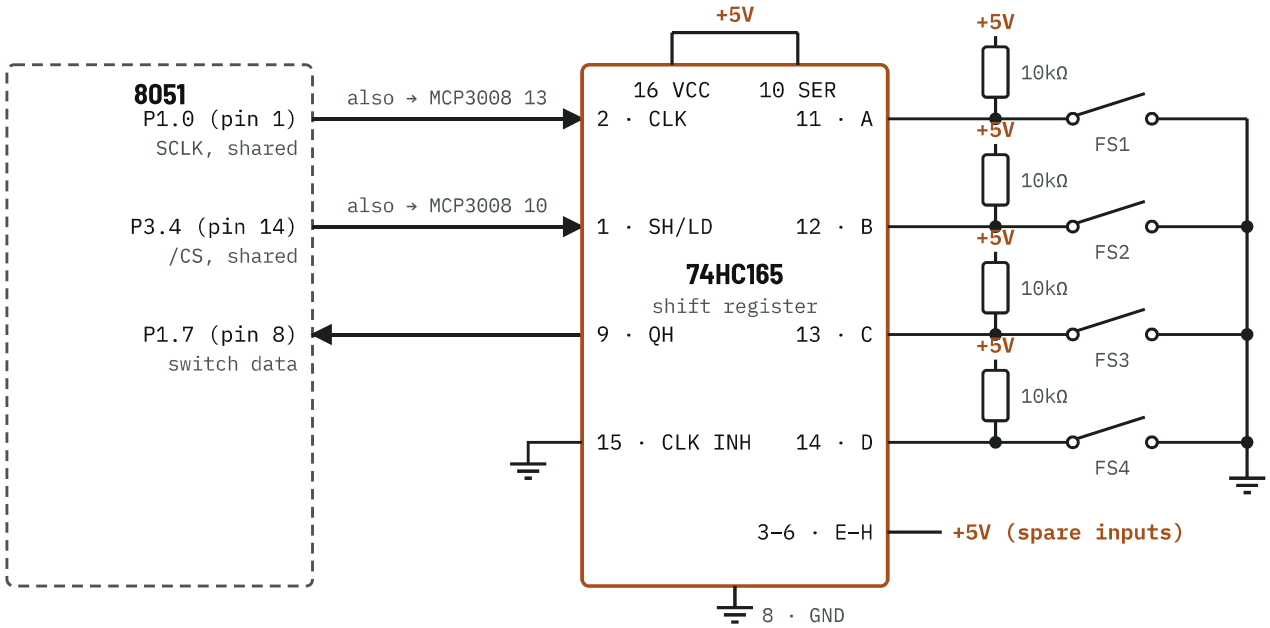
Pre-parts verification: transmit `0x55` continuously. Expected on TXD: 31.8 µs bit period. Measure with scope, or decode directly with the LogicPort.

`ACCEPT` Note On/Off received in MIDI-OX; Reaper synth track sounds.

M2 Debug port, footswitches

Owner: SW lead





HARDWARE

- MAX232 pins 7–10 unconnected per schematic rev 5. DB9 #2 pin 2 assignment: match DB9 #1.
- Alternative debug path: P3.5 → USB-TTL adapter RX, GND–GND. Eliminates DB9 #2, MAX232 ch2 and one USB-RS232.
- 74HC165: pin 16 → +5V, pin 8 → GND, 0.1 µF at pin 16. SER (10) → +5V, CLK INH (15) → GND, /QH (7) NC.
- Switch inputs A–D: 10 kΩ pull-up each (the '165 has no internal pull-ups); switch to GND, active low. Inputs E–H → +5V (expansion).

FIRMWARE

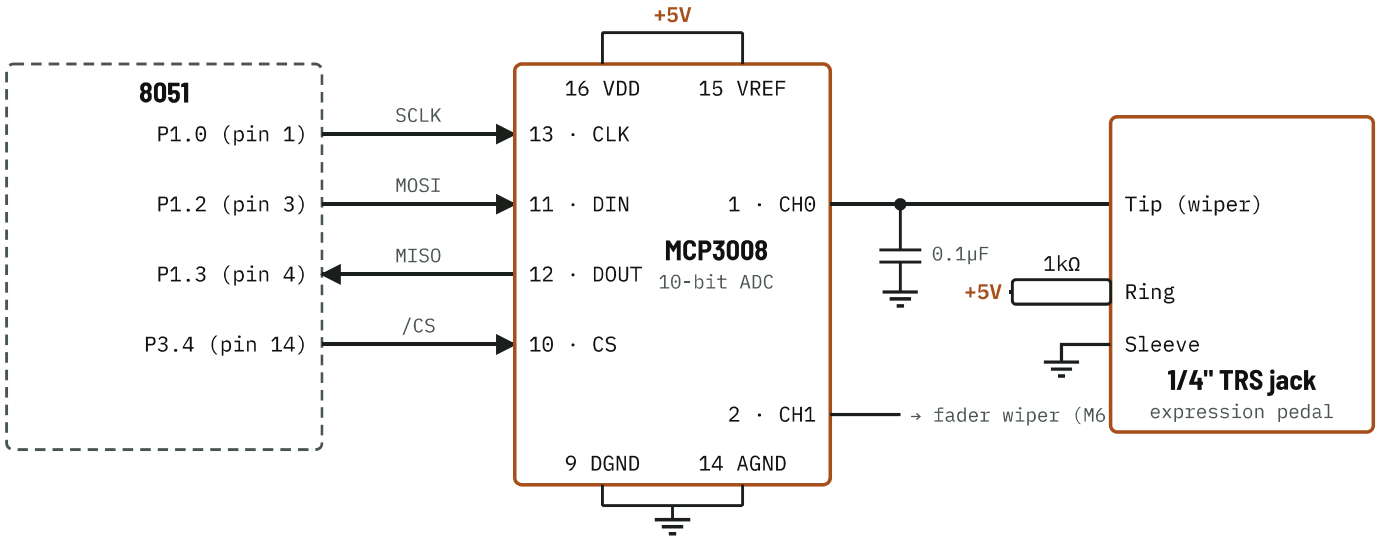
- Soft UART TX: Timer 0 mode 2, reload 0xA0 (96 × 1.085 µs = 104.17 µs = 1 bit @ 9600). Timer 0 at high interrupt priority.
- TX only: all three timers allocated.
- Debug throughput 960 B/s vs. MIDI IN up to 3,125 B/s. Keep messages short; no per-message line logging.
- Timer 1: 1 ms tick, reload 0xFC66.
- Switch read: pulse P3.4 low then high to load the '165 (the MCP3008 ignores a /CS pulse with no start bit). With P3.4 high, read P1.7 (input H), then 7 SCLK rising edges → G...A. The MCP3008 ignores SCLK while /CS is high.
- Debounce: state change accepted after 20 consecutive identical 1 ms samples.
- Mapping: FS1 → CC 20, 127 on press / 0 on release. Use CC 20–31 (undefined). Do not use CC 64 (sustain). Reaper: enable "input for control messages" on the device; bind via MIDI Learn. Mapping changes to Mackie Control transport notes in M6.

> **ACCEPT** FS1 toggles Reaper record; debug terminal logs the event.

M3 Expression pedal / ADC

Owner: AE lead

Pedal = potentiometer, 0–5 V on wiper. MCP3008 CH0 samples it; 8051 reads via bit-banged SPI. CH1 is reserved for the fader position (M6).



Drawing is not in physical pin order. Package: pins 1–8 = CH0–CH7 on one side, 9–16 on the other.

HARDWARE

- Bring-up with 10 kΩ pot: ends to +5V / GND, wiper to CH0.
- TRS wiper assignment varies by manufacturer. EX-P has a polarity switch. Verify with ohmmeter.
- 1 kΩ series on Ring supply. Limits fault current to 5 mA if a TS plug shorts Ring to Sleeve or pedal polarity is reversed.
- 0.1 µF decoupling on pin 16. 0.1 µF on CH0 for noise.

FIRMWARE

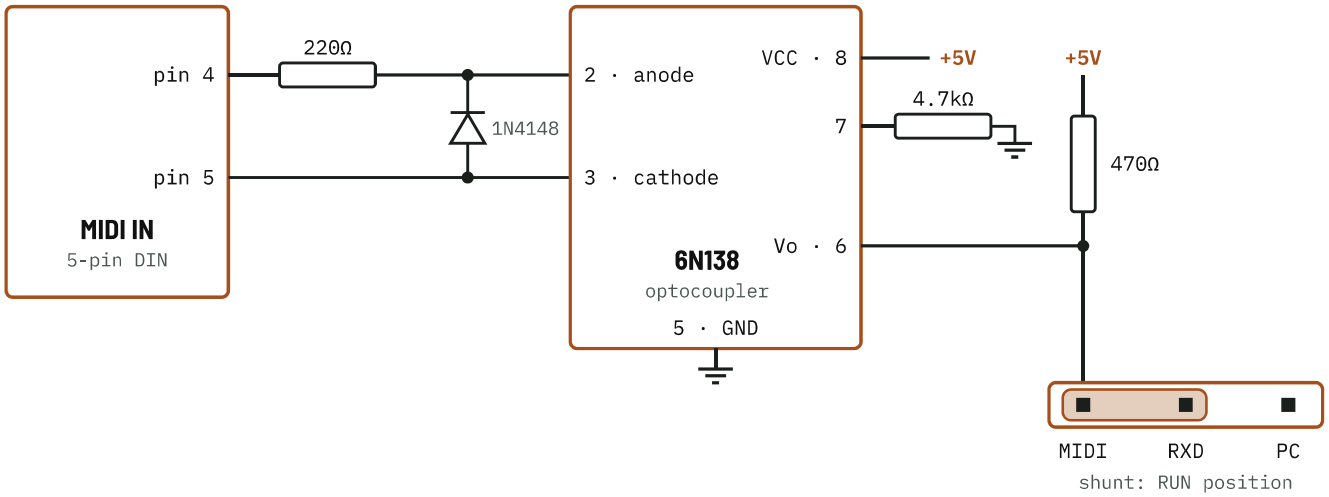
- ADC read sequence: /CS low, shift out 5 bits (start, SGL, D2, D1, D0). Then 1 sample clock, 1 null bit, 10 data bits MSB-first. Clock in 11 bits, discard the first. Omitting this yields a 1-bit shift.
- 8-sample moving average. Pedal sampled at 250 Hz (fader loop takes priority in M6).
- Calibration: 1 kΩ series limits full scale to ≈930 counts (10 kΩ pot). Store measured min/max; scale to 0–127.
- Hysteresis: transmit only when |reading − last sent| ≥ 4 counts. Prevents CC toggling at quantization boundaries.

> **ACCEPT** Full pedal travel maps to CC 0–127 in Reaper; zero messages at rest.

M4 MIDI IN, merge, filter

Owner: SW lead

Part 1: receive and parse MIDI IN. Part 2: merge MIDI IN with locally generated messages onto MIDI OUT; filter selected events (handout-listed feature). Primary firmware complexity item. MIDI IN also carries Reaper's fader feedback for M6.



HARDWARE

- RXD rework: remove MAX232 pin 12 (R1OUT) → P3.0 (pin 10). Wire R1OUT → header PC; header center → P3.0; 6N138 pin 6 → header MIDI.
- Shunt MIDI–RXD: run. Shunt RXD–PC: FLIP / PC serial. FLIP fails with shunt in RUN.
- During FLIP, bootloader traffic on TXD also appears on MIDI OUT. Disconnect the MIDI cable while flashing.
- DIN pin 2: no connection on MIDI IN (isolation).
- 1N4148 cathode to 6N138 pin 2. 0.1 µF on pin 8. Pins 1, 4: NC.

FIRMWARE

- RX ISR: received byte → 32-byte ring buffer. No processing in ISR.
- Parser state machine in main loop. ≥ 0x80 = status, < 0x80 = data.
- Running status: retain last channel status byte.
- Real-time (`0xF8` – `0xFF`): may interleave any message; process immediately; running status unchanged. System common (`0xF0` – `0xF7`): clears running status.
- Note On, velocity 0 ≡ Note Off.
- Merge: forward input to TX queue; insert local messages only at message boundaries. Separate running-status state for the output stream.
- Filter: lookup table. Defaults: Active Sensing (`0xFE`), aftertouch, selectable channels.
- Loop prevention: when Reaper drives MIDI IN, forwarded traffic returns to Reaper. The filter must drop control-surface feedback (fader pitch bend, LED notes) before merge.

> **ACCEPT** MIDI from an external source (keyboard, or Reaper output) passes through to Reaper during concurrent footswitch traffic. Zero dropped or corrupted messages; filtered events absent (MIDI-OX).

M5 LCD, presets, menu

Owner: SW lead

Firmware only. Depends on Lab 4 LCD installation. Uses bq4011 NVSRAM.

FIRMWARE

- Preset record: name, per-switch message, pedal CC.
- 4–8 presets at fixed XRAM 0x7F00–0x7FFF with checksum; defaults loaded on checksum failure. Region excluded from SDCC XRAM allocation.
- Edit menu entered by footswitch long-press.

LCD LAYOUT

- Line 1: preset name.
- Line 2: last message TX/RX, e.g. CC11 = 87, or fader position.
- Refresh ≤ 10 Hz.

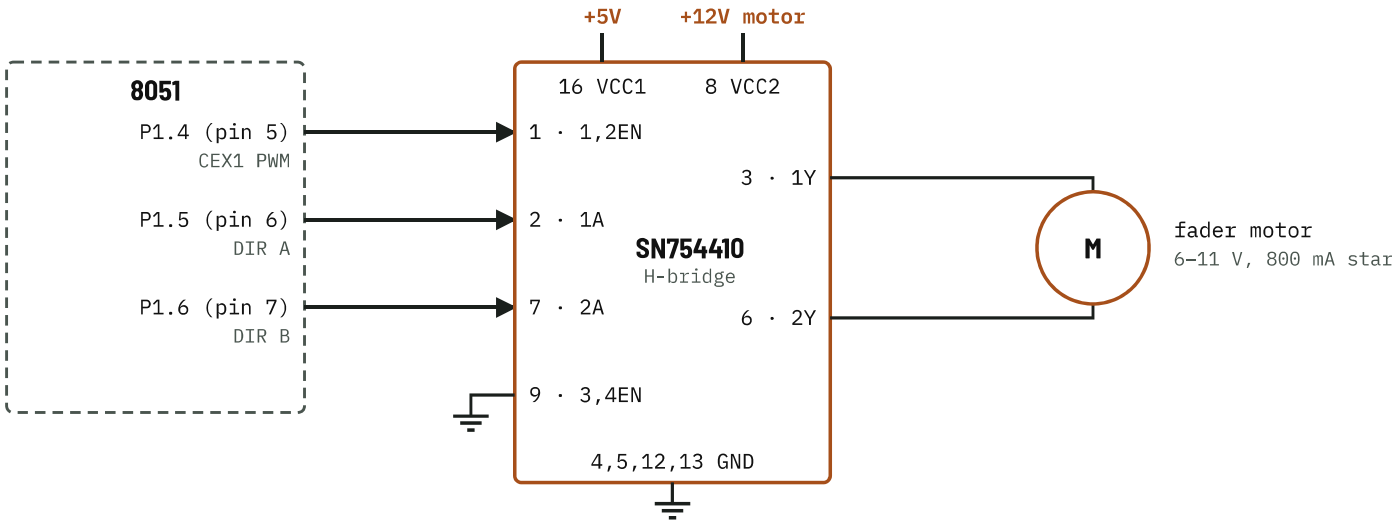
> **ACCEPT** Preset edited via footswitches only; persists across power cycle.

M6 Motorized fader

Owner: AE lead

Closed-loop position control. Fader = Reaper track volume. User moves fader → 8051 sends position. Reaper changes track or automation → Reaper sends position on MIDI IN → 8051 drives the motor to match. Implements the handout's "Digital Control System" topic inside the MIDI project.

SM100AN-B10K-L82T1H4-02 (MANUFACTURER DATA)	VALUE	DESIGN CONSEQUENCE
Track	10 kΩ ±20%, linear (B), 500 Ω residual, 0.5 W	Wiper never reaches the rails: ≈ 50 counts dead zone each end. Calibrate min/max.
Travel	100 ± 0.5 mm	≈ 0.1 mm per ADC count.
Motor rated / operating voltage	10 V DC / 6–11 V DC	5 V USB is below minimum. Do not use.
Starting current	800 mA	Driver ≥ 1 A continuous per channel.
Stall current (lever held)	400–800 mA	Firmware stall timeout required.
Lever speed	20 mm / 0.1 s (200 mm/s)	Full travel ≈ 0.5 s. 1 kHz loop = 0.2 mm per sample at full speed.
Sliding force (SM100AN)	10–80 gf	Hand-movable with motor off.
Code: 100A / N / L82T1 / H4	Vertical cylindrical motor, 100 mm · single gang · 8.2 mm T-lever with T hole · vertical, 2 PCB terminals each end, bracket pitch 96 mm	



HARDWARE

- Motor supply: USB-PD trigger module set to **12 V**, ≥ 1 A per fader. SN754410 output stages drop ≈ 2.6 V typ at 1 A (bipolar), so the motor sees ≈ 9.4 V under load and ≤ 11 V at light load. With a MOSFET driver (low drop), set the module to 9 V instead.
- SN754410: pin 16 → +5V (logic), pin 8 → +12 V (motor), pins 4/5/12/13 → GND (also heat sink). Pin 9 (3,4EN) and unused inputs 10, 15 → GND.
- Pin 8 decoupling: 100 µF electrolytic + 0.1 µF ceramic at the IC. 0.1 µF on pin 16.
- Grounds: PD module GND tied to board GND at one point near the SN754410. Motor current must not return through the 8051 ground path.
- Fader track: ends → +5V / GND (logic supply), wiper → MCP3008 CH1 with 0.1 µF to GND.
- Terminal identification: track ends read ≈ 10 kΩ ±20% between them; motor terminals read low resistance (tens of Ω).
- Reset state: port pins high → EN high, 1A = 2A high → both outputs high → motor braked. No motion at power-up.
- Driver dissipation at 0.8 A ≈ 2 W during motion. Acceptable for < 1 s moves; stall timeout bounds it.

FIRMWARE

- PCA module 1, 8-bit PWM. PCA clock $F_{CLK\ PERIPH}/2$ = 2.7648 MHz (X1) → 10.8 kHz PWM.
- Position: MCP3008 CH1 at 1 kHz on the Timer 1 tick.
- Control: PD on error = target − position → duty and direction. Deadband ±3 counts → brake (1A = 2A), then disable.
- Drive only on a new target from MIDI IN; release after arrival so the user can move the lever.
- Stall protection: drive cut if target not reached within 1 s (rated full travel ≈ 0.5 s).
- User movement: motor idle and position change beyond hysteresis → transmit position.
- Reaper link: Mackie Control Universal control surface on the USB-MIDI port. Fader 1 = 14-bit pitch bend, channel 1 (`0xE0` LSB MSB), both directions. 10-bit position scaled to 14-bit. Footswitch transport moves to MCU notes: Stop 0x5D, Play 0x5E, Record 0x5F.

> **ACCEPT** Track change in Reaper moves the fader to the stored volume within 1 s, ±1 mm, no oscillation. Moving the fader by hand changes Reaper track volume. Holding the lever during a move trips the stall timeout.

M7 Stretch (after M1–M6 accepted) Select one

TEMPO SYNC

MIDI Clock: 24 `0xF8` per quarter note. Derive BPM; LED on beat; tap tempo on a footswitch.

SECOND FADER

Second SM100AN on SN754410 channels 3/4 and MCP3008 CH2. Requires a second PCA PWM output and 2 direction pins: pin reallocation (Lab 4 reserve or switch E–H inputs repurposed). Motor supply ≥ 2 A.

Processor budget

11.0592 MHz, 12 clocks/machine cycle → 1.085 µs/cycle. Instructions: 1–2 cycles. Throughput < 1 MIPS.

TASK	RATE	BUDGET / NOTES
MIDI RX ISR	≤ 3,125/s (320 µs/byte)	~295 cycles between bytes. Buffer write only.
MIDI TX (merge)	≤ 3,125/s	Output bandwidth = input bandwidth. Under input saturation, local messages queue. Bounded queue; drop oldest pedal CC / fader update first.
Soft UART ISR (T0, high priority)	104 µs/bit while TX active	96 cycles/bit. ISR ≈ 20 cycles → ≈ 20% CPU during debug TX.
System tick (T1)	1 ms	Set loop flags. Tens of cycles.
Fader loop: ADC CH1 + PD (main loop)	1 kHz	≈ 200–250 µs per iteration → 20–25% CPU.
Pedal ADC CH0 + switch read	250 Hz	≈ 200 µs per 4 ms → 5% CPU.
Parser, merge, LCD, menu	main loop	Remaining time (≈ 50% with debug idle).

Priority: T0 high; UART, T1 low. UART has one byte time (320 µs) to read the receive buffer before overrun; T0 preemption is within margin. PCA PWM runs in hardware; no CPU cost.

Fallback

Milestones are independently demonstrable. If DIN hardware fails: route MIDI bytes out DB9 #1 at 38,400 baud (Timer 2 reload 0xFFF7, exact), then Hairless MIDI Serial Bridge → loopMIDI → Reaper. If motor control fails: the fader operates as an unpowered slider through the M3 ADC path.

LEVEL	DEMONSTRATED
Minimum (M1–M2)	Footswitch control of Reaper over MIDI.
Target (M1–M6)	+ expression pedal, MIDI IN with merge/filter, LCD, presets, motorized fader.
Stretch (M7)	+ tempo sync or second fader.

Schedule

Demo Fri 12/4 (course handout). Critical path risk: parts lead time. All parts ordered by Tue 10/20. Weeks 4–5 run two tracks in parallel.

WEEK	SW LEAD	AE LEAD
0 Sun 10/18	PDR slides submitted to Canvas.	
1 10/19–10/25	PDR Mon 10/19. Order parts by 10/20. Physically confirm pin allocation against schematic rev 5. Timer 2 baud bring-up + bit-period check. Install loopMIDI, MIDI-OX. Separate fader hardware from prior project; identify terminals. Begin M1 on parts arrival.	
2 10/26–11/1	M1 (joint), M2: debug port, 74HC165 switches	M1 (joint). Fader: measure track and motor; bench-run motor on 12 V through SN754410
3 11/2–11/8	M2 accepted. Begin M4 RXD rework	M3: ADC, pot, pedal. TA progress check
4 11/9–11/15	M4 pt 1: optocoupler, ring buffer, parser	M6 pt 1: driver wiring, PCA PWM, open-loop motion, CH1 position read
5 11/16–11/22	M4 pt 2: merge, filter, loop prevention	M6 pt 2: PD loop, stall timeout, MCU pitch-bend link
6 11/23–11/29	M5: LCD (requires Lab 4), presets, menu	M6 tuning; MCU transport notes for footswitches
7 11/30–12/4	Feature freeze Mon 11/30. Integration, M7 only if ahead. Enclosure/mounting. Demo video (graded, ≤ 8 min, 2 people). Demo Fri 12/4.	
Post 12/7, 12/9	Physical items Mon 12/7. Final report Wed 12/9.	

Ownership: SW lead — M2, M4, M5. AE lead — M3, M6. Joint — M1. Each member owns hardware and firmware items.

Power-on procedure

1. Wire with power off.
2. Verify IC orientation: pin 1 at notch/dot, counter-clockwise numbering (top view).
3. Continuity +5V to GND: must be open. Repeat for +12 V motor rail to GND and +12 V to +5V.
4. Continuity-check each new net against the schematic.
5. Apply logic power first; monitor for heat or odor; remove power on any anomaly. Apply motor power only after logic is up (motor held in brake by reset state).
6. Measure VCC–GND at each new IC: 5.0 V ±5%. Motor rail: 12 V ±5%.

Terms

UART / baud	Asynchronous serial; baud = bits/s. Both ends must match within ~2%.
Current loop	Signaling by presence/absence of ~5 mA current. Noise-immune over long cables.
Optocoupler	LED + photodetector in one package. Signal transfer with no electrical connection.
Buffer	Non-inverting gate with higher source/sink capability than an MCU pin.
Shift register (PISO)	Parallel-in, serial-out. Captures 8 inputs at once, then shifts them out one bit per clock.
H-bridge	Four switches around a motor; selects current direction. Enable pin PWM sets average voltage.
PWM	Fixed-frequency square wave with variable duty cycle. Average motor voltage = duty × supply.
PD control	Drive = Kp × error + Kd × rate of change of error. The derivative term brakes before the target to prevent overshoot.
Pull-up	Resistor holding an undriven line high. Switch to GND overrides it.
Decoupling cap	0.1 µF at each IC supply pin. Supplies switching transients locally.
ADC	Analog-to-digital converter. MCP3008: 10-bit, $0–V_{REF} \rightarrow 0–1023$.
SPI	Synchronous serial: SCLK, MOSI, MISO, /CS. Bit-banged here.
Debounce	Contact bounce lasts ms; accept state only after it is stable for a set time.
Running status	MIDI optimization: repeated status byte omitted. Receiver retains last status.
Mackie Control (MCU)	De facto DAW control-surface protocol over MIDI. Faders as pitch bend, buttons as notes.
Jumper / shunt	Removable link across two header pins. Manual signal routing.

AI disclosure

Plan text drafted with Claude (AI) during project brainstorming; contains no source code. PDR and final-project implementation: no AI assistance. Verify all pin assignments and values against datasheets and the physical board before wiring.